

Ethics of AI #4

ECE Paris – Ing5 Data&IA



Farès Fadili

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Overview of the course

5 two-hours classes, including two practical works and one bonus-graded project :

1. **Culture of AI Ethics** : lecture and group presentations
2. **Biases** : lecture and practical work
3. **Fine-tuning** : lecture and practical work
4. **XAI & Ethics by design** : lecture ; **Project** : start of group work 👉
5. **Project** : group presentations

Only one “real” grade : **written exam (MCQ)** probably on December 12nd



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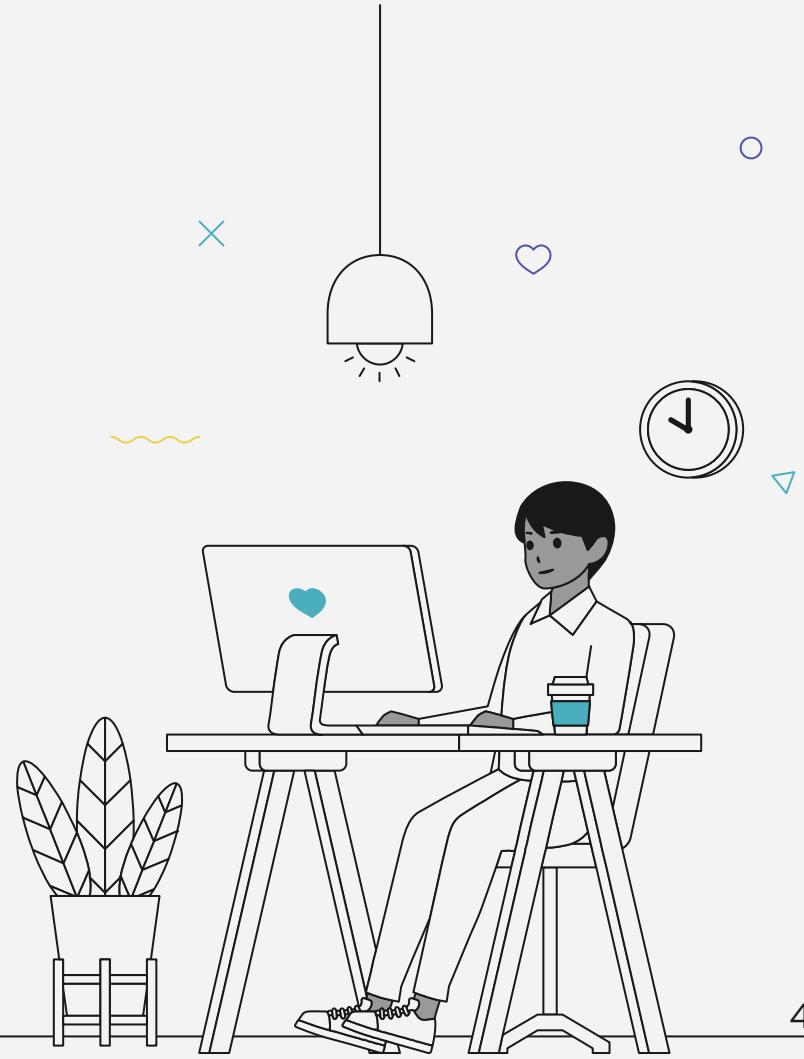
XAI : practice

How to actually
implement XAI ?

01

Ethics by design

Definition, roots, notable methods





1.1. Definition

The Concept: **Integrate** ethical considerations **from the beginning** of the product life cycle, not at the end.

The paradigm shift:

Before (Audit): "Is our AI ethical?" (Too late).

Now (Design): "How can we design ethical AI?"

Analogy: The **seatbelt** (designed with the car, not after the accident).





1.2. Roots : privacy by design (½)



1. Proactive, not reactive; preventive, not curative : Anticipate and prevent privacy risks before they occur.

2. Privacy by Default : Users don't have to do anything to be protected. The most secure setting is the default setting (e.g., the “newsletter” box is not pre-checked).

3. Privacy Embedded : Protection is not a “module” added at the end. It is part of the core architecture of the system from day one.

4. Full functionality – Positive-sum game : Avoid false dilemmas (e.g., “it's either high-performance or private”). Design so that both objectives (usefulness and privacy) are achieved.





1.2. Roots : privacy by design (2/2)



5. End-to-end security – Protection throughout the entire lifecycle : Secure data at every stage: from collection (encryption) to storage, right through to permanent deletion.

6. Visibility and transparency – Remain open : Users must be able to understand, clearly and honestly, how their data is collected, used, and shared.

7. Respect for user privacy – Staying user-centered : Put the user's interests and control at the center of the design (e.g., give them access to their data and the right to delete it).





1.3. Notable methods

Mapping stakeholders:

Who is **impacted**? **Beyond** the **end** user. (E.g., people not targeted by the service, society, the environment, etc.).

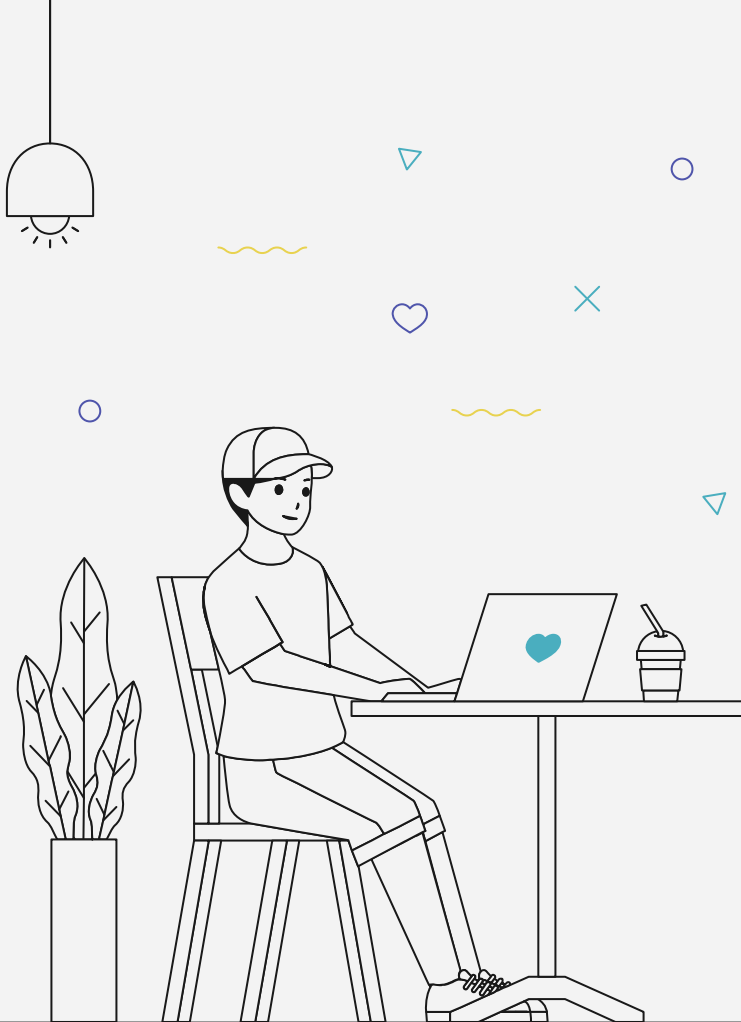
Ethical Red Teaming:

Your mission: **Think** like the **“enemy”** How can your AI be hijacked? (Jailbreak it, force it to be biased, make it reveal data, etc.).



02 XAI : theory

Definition, approaches, tools, limitations





2.1. Definition and issues

XAI : e**X**plainable **AI**

A set of **techniques** that enable humans to **understand** the decisions made by an AI **model**.

Why is this crucial? (*Reminder CI*)

Trust & Adoption: A doctor will not use a diagnosis that they do not understand.

Debugging & Robustness: For us (engineers): is the model looking at the right thing?

Legality: The AI Act requires it for high-risk systems (right to an explanation).





2.2. Main approaches : transparent models



The model is simple and **readable** by nature.

Examples: Linear regression (the coefficients can be **read**), decision trees (the path can be **followed**).

Advantage: The explanation **is** the model.

Limitation: Often **less effective** on complex problems (e.g., images, text).





2.2. Main approaches : post-hoc explanations



The model is a “**black box**” (e.g., deep learning).

A **second** simple model is used to **explain** the **first**.

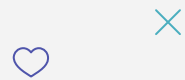
LIME : Local Interpretable Model Agnostic Explanation

SHAP : SHapley Additive exPlanations

Advantage: Allows the use of **high-performance** models.

Limitation: The explanation is an **approximation**.





2.3. Post-Hoc XAI Tools : use case



An AI model **predicts** whether someone is going to purchase a product.

The sample : "Harold - Age : 25, Income: €40,000/year, Has seen the ad : 1 (Yes)"

The Black Box (AI): "Prediction: YES"

In fact, the loan **model** does **not simply say "Yes"** for Harold. It calculates a **probability** score, for example 90% (chance of being accepted). As this score is above 50%, the firm **translates** it as "Yes."





2.3. Post-Hoc XAI Tools : LIME (1/3)

"What variables does this decision primarily rely on ?"

Step 1: Create data "around" the point to be explained.

LIME generates **slightly** modified versions of this example:

<u>Example</u>	<u>Age</u>	<u>Income</u>	<u>Ad viewed</u>	<u>Model prediction</u>
Original	25	40	1	0.90
Ver 1	25	40	0	0.30
Ver 2	25	35	1	0.70
Ver 3	35	40	1	0.85
Var 4	45	20	0	0.10

LIME notes how the probability changes when each variable is changed.





2.3. Post-Hoc XAI Tools : LIME (2/3)

Step 2: Give **more importance** to similar examples

LIME assigns a **weight** to each example:

The **closer** an example is to the original case, the **more** it **counts** in the explanation.

Thus, Ver 1, 2, and 3 (close to the base example) will have more weight than Ver 4.

Step 3: Learn a simple (local) model

LIME then builds a **small linear model** (e.g., a simple regression) on these weighted points.

This model seeks to mimic the “local” **relationship** between the variables and the prediction.

The local regression could give something like:

Local prediction = 0.1 + 0.05 × Age + 0.2 × Income + 0.6 × Ad viewed





2.3. Post-Hoc XAI Tools : LIME (3/3)

Step 4: Read the coefficients

The local model coefficients tell you directly which variables have the **greatest influence** on the prediction:

Variable	Local weight	Interpretation
Ad viewed	+0.6	Having seen the ad greatly increases the probability
Income	+0.2	A high income helps a little
Age	+0.05	Being young helps slightly

So LIME **concludes**:

“The high probability (0.9) is **mainly due** to the variable Ad seen”





2.3. Post-Hoc XAI Tools : SHAP (¼)



"How much of this decision is due to each variable?"

SHAP is based on Shapley's **game theory**, originally used to distribute winnings fairly among several players who collaborated on a result.

In the case of an AI model:

The **players** are the **variables** (age, income, ad viewed, etc.).

The **payoff** is the **model's prediction** (e.g., probability of purchase = 0.9).

Here, the model predicts that the person will buy with a probability of **0.9**.

Without any information (average model value), the base probability is **0.5**.

We therefore need to **explain** the **+0.4 difference** ($0.9 - 0.5$).



2.3. Post-Hoc XAI Tools : SHAP (2/4)

STEP 1 : Consider all possible combinations of variables (2^n with n the nb of variables)

Empty group: no information (base = 0.5)

{Age}

{Income}

{Ad viewed}

{Age, Income}

{Age, Ad viewed}

{Income, Ad viewed}

{Age, Income, Ad viewed}

For each group, SHAP **calculates** the model's prediction based **only** on these variables (the **targeted** variable uses the **use case value** while the **others** are “**averaged**” or “**masked**”).





2.3. Post-Hoc XAI Tools : SHAP ($\frac{3}{4}$)



STEP 2: Calculate the contribution “margin” of each variable

Next, for each variable, SHAP looks at how much value it adds when added to a group.

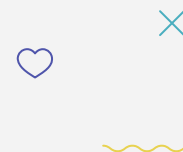
Example for the variable “Ad viewed”:

Group before	Prediction without Ad	Prediction with Ad	Gain contributed by Ad
{}	0.5	0.7	+0.2
{Age}	0.55	0.75	+0.2
{Income}	0.6	0.9	+0.3
{Age, Income}	0.65	0.95	+0.3

SHAP then averages these gains → SHAP value of “Ad viewed”.

SHAP value(Ad) = $(0.2+0.2+0.3+0.3)/4=0.25$





2.3. Post-Hoc XAI Tools : SHAP (4/4)

STEP 3: Repeat for each variable

Same calculation for:

Age $\rightarrow +0.05$

Income $\rightarrow +0.1$

(Ad viewed $\rightarrow +0.25$)

Total = $0.05 + 0.1 + 0.25 = 0.4 \rightarrow$ consistent with the difference between 0.9 and 0.5.

So SHAP **concludes**:

"The baseline prediction was 0.5. To arrive at the final score of 0.9 : the variable Ad seen contributed +0.25 points, Income +0.10 and Age +0.05"





2.3. Post-Hoc XAI Tools : benchmark



The main advantage of **LIME** is its **speed** and **simplicity**.

Because it only calculates a local approximation (by testing random “neighbors”), it is much faster than **SHAP**, which is **accurate** but very **slow**.

LIME's **explanation** is also **easier** for non-experts to understand: it provides a simple list of the “most important factors” (e.g., “Rejected because of fixed-term contract”).

This makes **SHAP** the ideal tool for providing “**live**” **explanations to end users**, while SHAP is better suited to **in-depth technical audits**.





2.4. Limitations

An explanation is **not a justification**

Ex : AI can explain that it refused a loan because “the zip code is the no.1 factor.”

The explanation is **transparent**, but the decision is **unfair**.

Conclusion: XAI is a **debugging** and **questioning** tool, not a “certificate of non-bias.”



03

XAI : practice

How to actually implement XAI ?





3. Practical work

Use case: AI-powered decision to grant or deny a bank loan

Model: Random forest

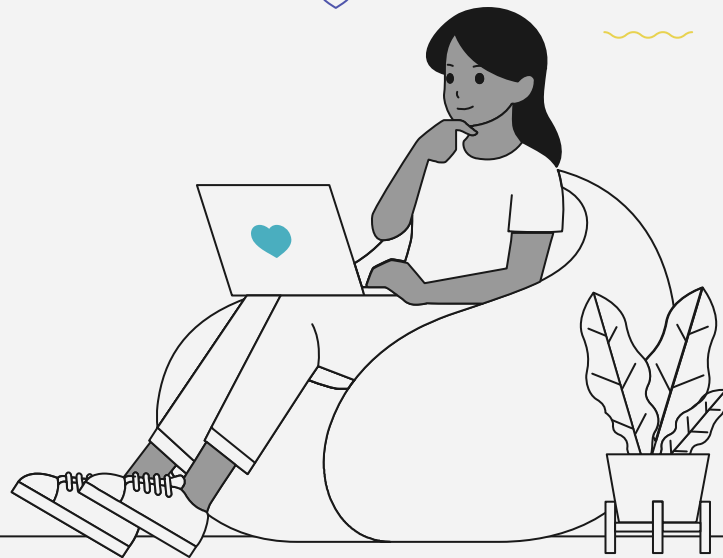
Goal: Explain the model's decisions regarding a set customer, using LIME and SHAP



04

Project

Review of the course





4.1. Guidelines



Goal : make ethical (by design) a classic AI workflow

Steps

- Find a **real-life dataset** (on Kaggle & co), enlight its **biases** and reduce them if you decide to implement an unbiased prediction but not for a realistic prediction (slides 27-28)
- Get a handleable **LLM**, **adapt** it by the three methods and **benchmark** them in order to save the most appropriate one
- Use it as a **solution** for a relevant problem regarding your dataset
- **Explain** its decisions thanks to both XAI tools

Teams of 3-4 students (**10 teams** per group), **10-min presentations** at the next class

Notebook and **slides** to upload on Boostcamp before the presentation

Up to 3 bonus points on the written exam grade





4.2.1. Examples : realistic prediction



Dataset : Forward market values prediction ; features : statistics, country, market value, etc.

Solution/title : LLM-powered forward market values prediction

Bias : Despite scoring less goals, forwards from very few countries are always more valued than others

Bias reduction : None (*unless you want to turn it into a decision-making solution*)

Final demo (asking the specialized LLM) : What is the market value of a 23-years old moroccan forward who scored 6 goals and provided 1 assist in 9 games this season with Lille ? And of a brazilian one with nearly the same features ? Why ? (cf. XAI)





4.2.2. Examples : unbiased prediction



Dataset : College first-year admissions ; features : grades, location, admitted (0/1), etc.

Solution/title : LLM-powered college first-year admissions decision-making

Bias : Despite having same baccalauréat grades, suburbs students are widely less admitted

Bias reduction :

- **Shortening/altering** the dataset to train your model more fairly
- And/or keep your dataset unchanged and **configure** your model

Final demo (asking the specialized LLM) : Between Jonas (18/20 and from a remote area) and Bartosz (17.5/20 and from the capital), who was before the biases reduction more likely to be admitted and now is ? Why ? (cf. XAI)





**Thanks for your
attention.**

